

Enhancing Operational Efficiency in a Multispecialty Hospital

Name: Fadhli Hatta

Date: 04/02/2026



Table of Contents

- Executive Summary
- Introduction
- Business Objectives

Methodology

- Requirements Gathering (BRD and RTM)
- Stakeholder Analysis and Engagement Plan
- Scope Management Plan
- Data Analysis
- Data Visualization
- Process Mapping
- Risk Assessment Plan
- Risk Mitigation Plan

Findings and Recommendations

- Key Findings
- Key Recommendations
- Conclusion

Appendix

Executive Summary

Problem:

1. Manual and fragmented appointment and patient flow processes lead to long waiting times, inefficient resource utilization, and frequent interdepartmental communication issues.

Key insights:

1. Inefficiencies stem from manual scheduling, verification, and handoffs across departments.
2. Limited real-time system integration prevents optimal use of clinical and administrative resources.

Recommended actions:

1. Implement an integrated digital appointment and check-in system.
2. Enable real-time data integration across clinical, pharmacy, and billing units.
3. Monitor operational performance to improve efficiency and service quality.

Introduction

Summarize the opportunity, describe the approach, and outline the key questions or hypotheses to be analyzed in 6 bullet points.

Opportunity:

1. Reduce patient waiting times and administrative inefficiencies
2. Improve utilization of clinical and administrative resources

Approach:

1. Analyze current patient flow to identify bottlenecks
2. Design an integrated, digital future-state process

Key questions/hypotheses:

1. Can digital scheduling and check-in reduce delays and errors?
2. Will real-time data integration improve coordination and resource use?

Business Objectives

Areas of improvement in 6 bullet points:

1. Reduce average patient waiting time by at least 30% within 3 months of system implementation.
2. Achieve $\geq 95\%$ successful online bookings with zero double bookings within the first release.
3. Increase doctor resource utilization to 70–85% within 3 months through optimized scheduling.
4. Enable real-time data integration across all departments (admin, clinical, pharmacy, billing) at go-live.
5. Improve patient satisfaction score by at least 20% within 6 months after implementation.
6. Maintain system uptime of $\geq 99.5\%$ while ensuring full regulatory and data security compliance.



Methodology

Requirements Gathering: Business Requirement Document (BRD)

Problem statement:

The hospital currently faces challenges in appointment scheduling, resource management, and communication across departments. Patients experience long wait times and unclear communication, while staff struggle with overbooking, outdated systems, and limited system integration.

Key requirements to improve operational efficiency:

The requirements focus on efficiency, integration, reliability, and scalability to improve hospital operations and patient experience.

Requirements Gathering: Business Requirement Document (BRD)

Constraints:

- Limited IT budget requiring phased implementation
- Dependency on existing legacy systems
- Regulatory and data privacy compliance requirements
- Limited change capacity for operational staff

Acceptance criteria:

- $\geq 95\%$ successful online bookings with no double booking
- $\geq 30\%$ reduction in average patient waiting time
- Real-time data integration across all departments
- System uptime $\geq 99.5\%$ and regulatory compliant

Requirements Gathering: Requirement Traceability Matrix (RTM)

Requirement ID	Requirement Description	Priority (MoSCoW)	Stakeholder(s)	Project Objective	Related Data File	Status
FR1	Online appointment booking with real-time availability	Must Have	Patients, Admin Staff	Reduce wait times and scheduling errors	Appointment_data.csv	
FR2	Automated appointment reminders and notifications	Should Have	Patients, Admin Staff	Reduce no-show rate and last-minute cancellations	Appointment_data.csv	
FR3	Integrated patient records accessible across departments	Must Have	Doctors, Nurses, Admin	Reduce inter-department miscommunication	N/A	
FR4	Resource dashboard for administrators to monitor operations including rooms, equipment, and staff	Must Have	Hospital Management, IT	Optimize resource allocation and capacity planning	resource_data.csv	
FR1	Online appointment booking with real-time availability	Must Have	Patients, Admin Staff	Reduce wait times and scheduling errors	Appointment_data.csv	

Requirements Gathering: Requirement Traceability Matrix (RTM) (2)

Requirement ID	Requirement Description	Priority (MoSCoW)	Stakeholder(s)	Project Objective	Related Data File	Status
NFR1	System uptime of at least 99.5%	Must Have	Hospital Management	Ensure service availability and operational continuity	N/A	
NFR2	Secure data storage compliant with healthcare regulations	Must Have	Hospital Management, IT	Ensure regulatory compliance and data protection	N/A	
NFR3	User-friendly interfaces for patients and staff	Should Have	Patients, Staff	Improve adoption and reduce operational friction	feedback_data.csv	

Stakeholder Analysis and Engagement Plan

Stakeholders and Influence:

Patients – Low

Doctors – High

Nurses – Medium

Administrative Staff – Medium

IT Teams – High

Hospital Management - High

Stakeholder Analysis and Engagement Plan

Stakeholder engagement strategies:

Engage stakeholders based on influence and interest through targeted interviews, workshops, and user feedback to validate requirements and process changes.

Stakeholder communication strategies:

Provide regular, role-specific updates via meetings, reports, and surveys to ensure transparency, alignment, and timely decision-making.

Scope Management Plan

In-scope activities:

1. Integration of scheduling and record management systems
2. Real-time appointment notifications via email/SMS
3. Improved visibility of resource availability
4. Data analytics for patient and resource trends

Out-of-scope activities:

1. Replacement of all clinical equipment
2. Redesign of hospital physical infrastructure

Scope Management Plan

Assumptions:

- Strong management support for system integration and change
- Staff are trained and willing to adopt the new system
- Patient and operational data are accurate and usable
- Patients have access to basic digital tools
- Interdepartmental workflows can be standardized

Constraints:

- Limited IT budget requiring phased implementation
- Dependency on existing legacy systems
- Regulatory and data privacy compliance requirements
- Limited staff availability for training and change
- Fixed project timeline aligned with hospital operations

Scope Management Plan

Phases in the Work Breakdown Structure (WBS):

WBS ID	Task Name	Task Description	Milestone
1.0	Project Governance & Control	Governed, compliant, and controlled project execution	Project Manager
2.0	Requirements & Scope Baseline	Validated and traceable solution scope	Business Analyst
3.0	Appointment Coordination Capability	Safe and synchronized scheduling capability	IT Team
4.0	Resource Visibility & Utilization	Real-time resource intelligence	IT Team
5.0	Patient Communication & Notification	Automated operational communication	IT Team
6.0	HIS Integration & Data Governance	Safe and reliable interoperability	Integration Team
7.0	Clinical Workflow Safety & Continuity	Clinically safe and uninterrupted operations	Clinical Lead

Scope Management Plan

Phases in the Work Breakdown Structure (WBS) (2):

WBS ID	Task Name	Task Description	Milestone
8.0	Operational Analytics & Intelligence	Operational performance intelligence	Data Team
9.0	Security, Privacy & Compliance	Secure and compliant system	IT Security
10.0	Organizational Readiness & Adoption	Organization capable of operating system	Business Analyst
11.0	Deployment & Operational Transition	Stable production-ready system	IT Team
12.0	Benefits Realization & Validation	Validated operational improvements	PMO + Business

Scope Management Plan

Scope Governance Principles:

- All scope elements must trace to Business Objectives → Requirements → Deliverables → Acceptance
- Scope defined using capability-driven and deliverable-oriented Work Breakdown Structure (WBS)
- Clinical safety and operational continuity are mandatory acceptance dimensions
- Data governance and interoperability standards must be enforced
- Scope baseline cannot change without formal Change Control Board (CCB) approval
- Bidirectional traceability must be maintained at all times

Process Mapping

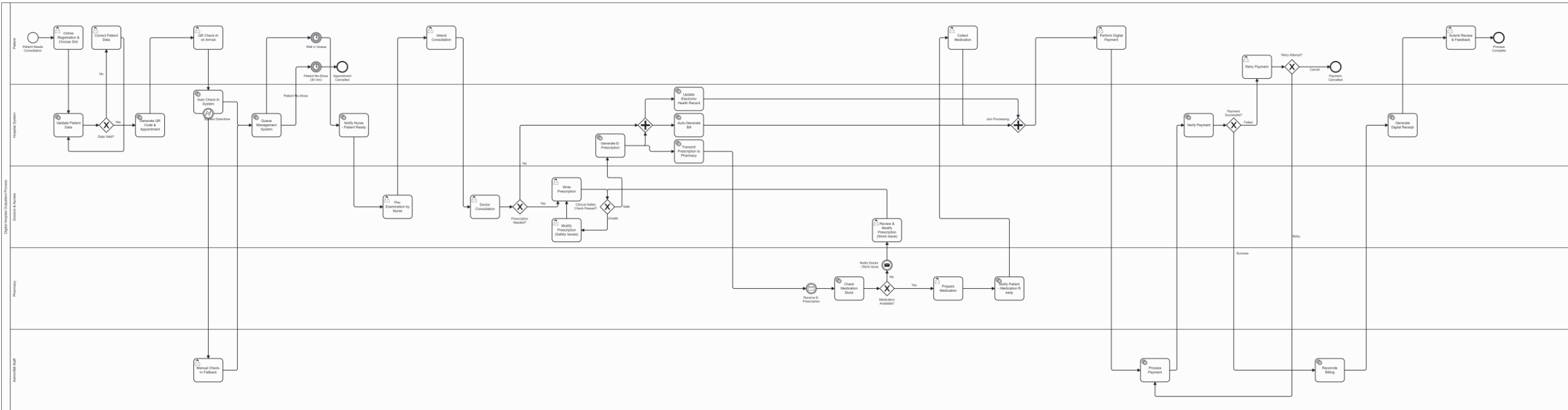
Process	As-Is Model	To-Be Model
Registration & Booking	Patient submits form → partial system validation → errors corrected manually via admin → booking verified manually → queue dependency begins	Patient submits form with real-time slot → system auto-validates → patient self-corrects errors → booking auto-confirmed with QR → scheduled flow established
Arrival & Check-In	Patient reports to admin → manual booking verification → manual queue waiting	Patient scans QR → system auto check-in → visit activated automatically → time-aligned scheduling reduces queue
Consultation & Clinical Recording	Consultation performed → documentation limited → downstream processes handled manually	Consultation auto-recorded in EHR → integrated clinical workflow → digital decision support

Process Mapping (2)

Process	As-Is Model	To-Be Model
Prescription & Pharmacy	Manual prescription → patient instructed to visit pharmacy → pharmacy manually validates data → medication prepared → inventory not integrated	E-prescription auto-generated → transmitted digitally → pharmacy receives verified data → faster preparation → stock auto-updated
Billing & Payment	Patient goes to billing → manual billing calculation → manual payment → no structured failure handling	Billing auto-generated → digital payment → automatic verification → retry mechanism for failure
Closure & Post-Service	Limited record update → manual / inconsistent receipt → no structured feedback capture	EHR updated automatically → e-receipt generated → digital patient feedback captured

Advanced Process Mapping

Detailed workflow using the advanced BPMN model:



Advanced Process Mapping

Stakeholder responsibility using the Swimlane diagram:

Swimlane (Stakeholders)	Task / Activity	Description
Patient	Registration, Check-In, Consultation, Medication Collection, Payment, Feedback	Initiates registration, completes check-in, attends consultation, collects medication, performs payment, and provides feedback, acting as both the starting and ending actor of the process.
Hospital System	Validation, Appointment & Queue Management, EHR Update, E-Prescription, Billing & Payment Verification	Acts as the central orchestrator by validating data, managing scheduling, recording clinical data, generating prescriptions and billing, verifying payments, and ensuring end-to-end process integration.
Doctors & Medical Staff	Examination, Diagnosis, Prescription Decision, Safety Review	Performs clinical evaluation, determines treatment and prescription, and ensures medical accuracy and patient safety.

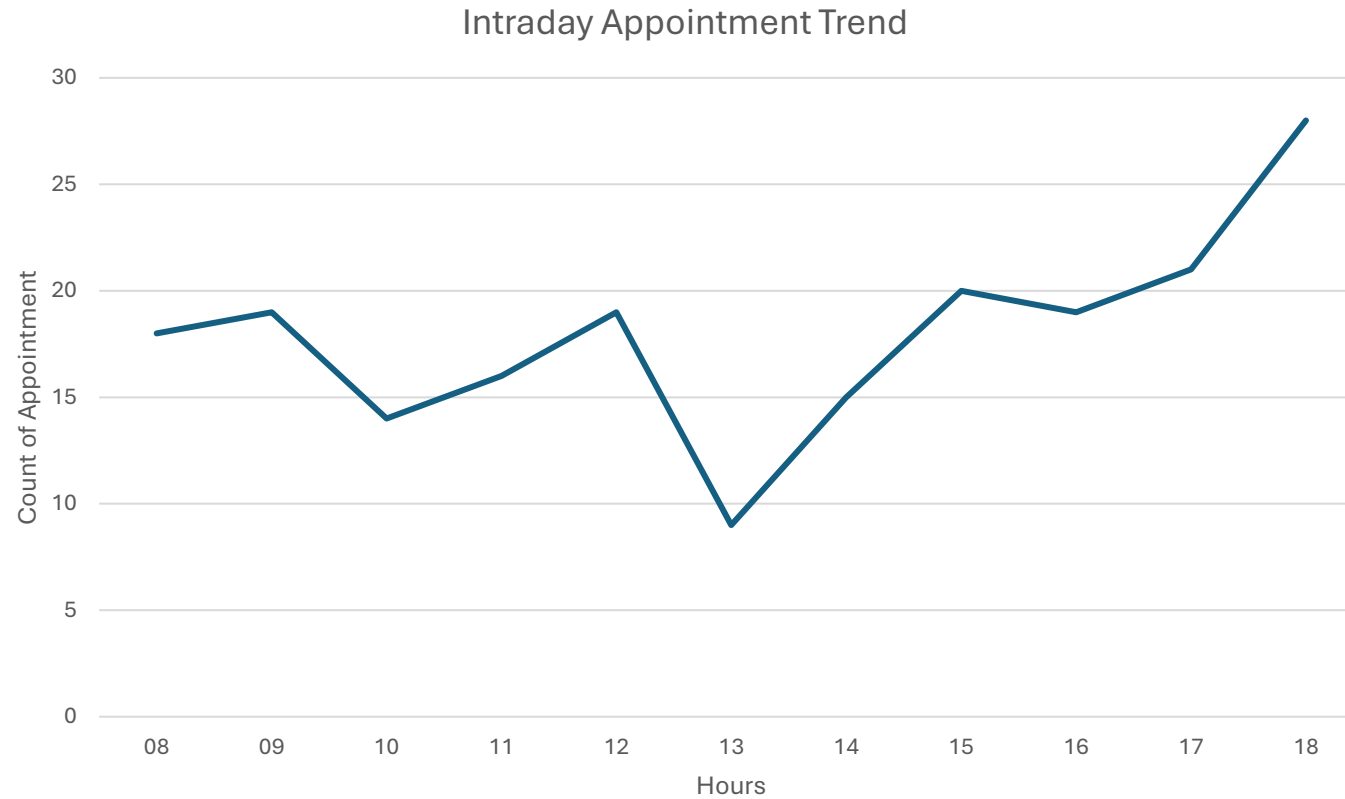
Advanced Process Mapping

Stakeholder responsibility using the Swimlane diagram (2):

Swimlane (Stakeholders)	Task / Activity	Description
Nurse	Initial Assessment, Patient Preparation, Coordination	Conducts preliminary assessment, prepares patient, and ensures readiness for consultation.
Pharmacy	Receive Prescription, Stock Check, Medication Preparation, Inventory Update	Receives verified prescription, prepares medication, dispenses to patient, and maintains accurate stock records.
Admin / Billing Staff	Exception Handling, Manual Support, Billing Oversight	Handles exception cases, supports manual fallback when needed, and ensures billing and financial accuracy.

Data Analysis

Trends using a Pivot Table:



Data Analysis

Trends analyzed from the Pivot Table:

1. Two clear peak periods (bimodal pattern)

Morning peak: around **9:00 AM – 10:00 AM**

Evening peak: strongest at **6:00 PM – 6:30 PM (highest overall)**

→ Indicates demand concentrated before work hours and after typical workday.

2. Midday dip (low demand window)

Lowest activity around **10:30 AM – 1:30 PM**

Especially weak near **11:00 AM – 12:00 PM** and **~1:30 PM**

→ Suggests underutilized capacity during late morning to early afternoon.

3. Gradual rise in afternoon demand

From **2:30 PM onward**, appointments steadily increase

Peaks again toward late afternoon and early evening

→ Indicates second demand wave after midday lull.

Data Analysis

Trends analyzed from the Pivot Table (2):

4. Demand skew toward later hours

Late afternoon and evening show **more consistent high volume** than early morning

→ Possible after-work patient preference.

5. High variability during morning hours

Fluctuations between **6:30 AM – 10:30 AM**

→ May reflect mixed walk-in vs scheduled behavior or uneven slot distribution.

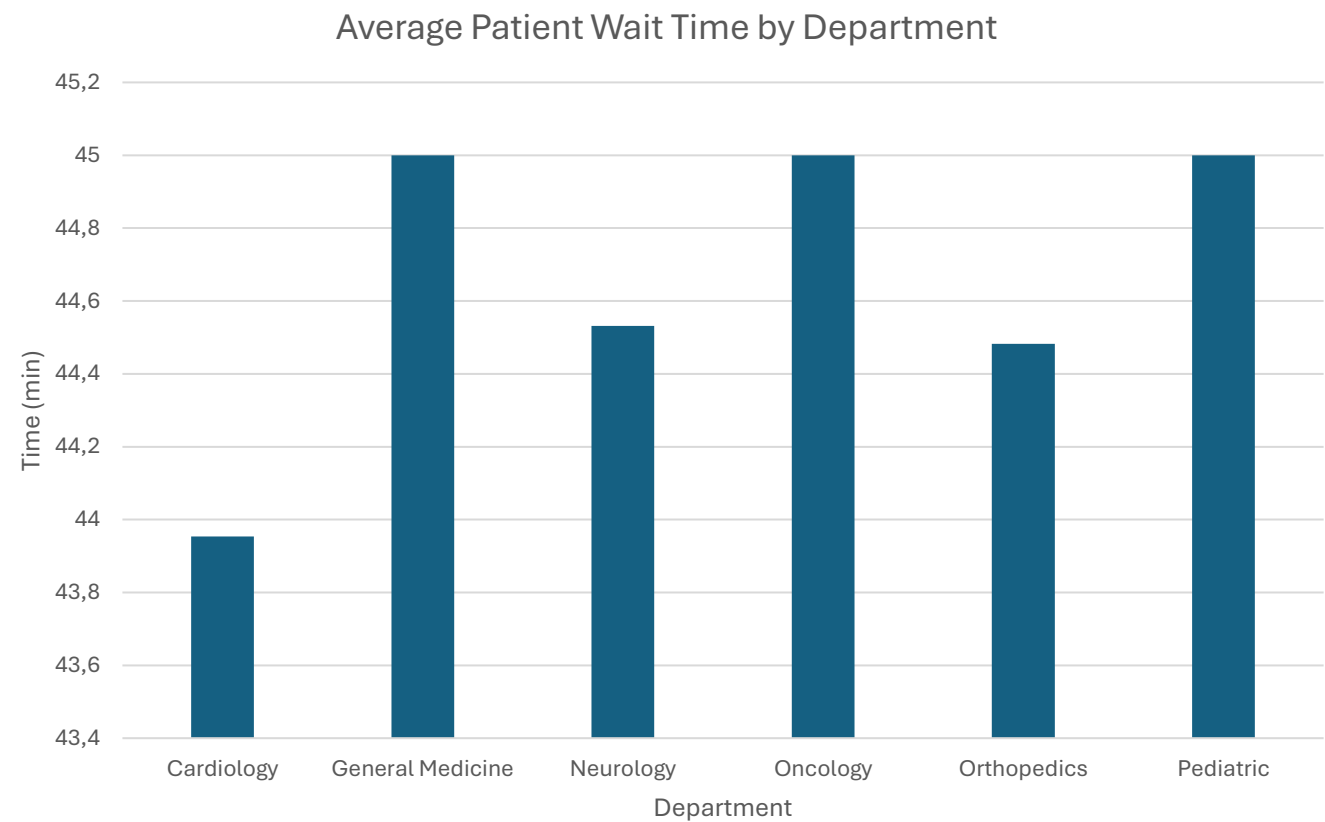
Data Analysis

Key insights:

1. Appointment demand shows two peak periods: morning (9–10 AM) and strongest in the evening (6–6:30 PM)
2. Midday (10:30 AM – 1:30 PM) shows the lowest activity, indicating underutilized capacity
3. Demand gradually increases again from mid-afternoon toward evening
4. Evening hours show more consistent high volume than morning
5. Uneven distribution suggests opportunity for load balancing and schedule optimization

Data Visualization

Average patient wait time using a horizontal bar chart:



Data Visualization

Bar chart highlighting overused and underutilized resources:

- Overutilized departments:**

General Medicine, Oncology, and Pediatric show the **highest wait times (~45 min)**, indicating resource strain and potential overload.

- Moderately utilized:**

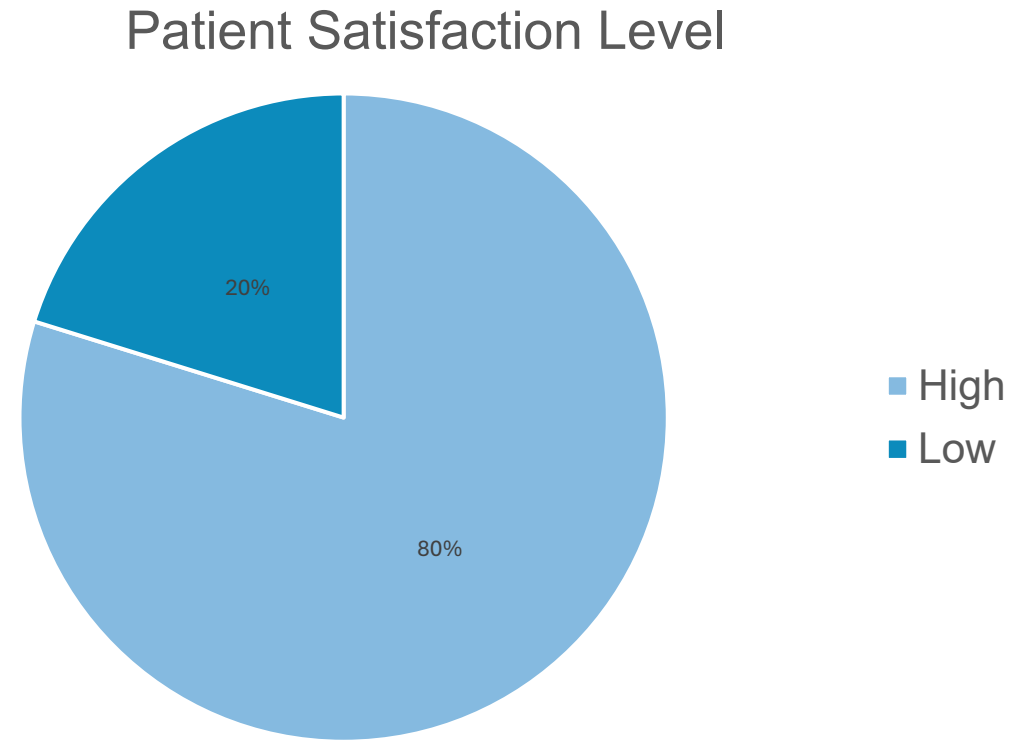
Neurology and Orthopedics show **slightly lower wait times (~44.5 min)**, suggesting near-capacity operations.

- Underutilized department:**

Cardiology shows the **lowest wait time (~44 min)**, indicating relatively lower demand or excess capacity.

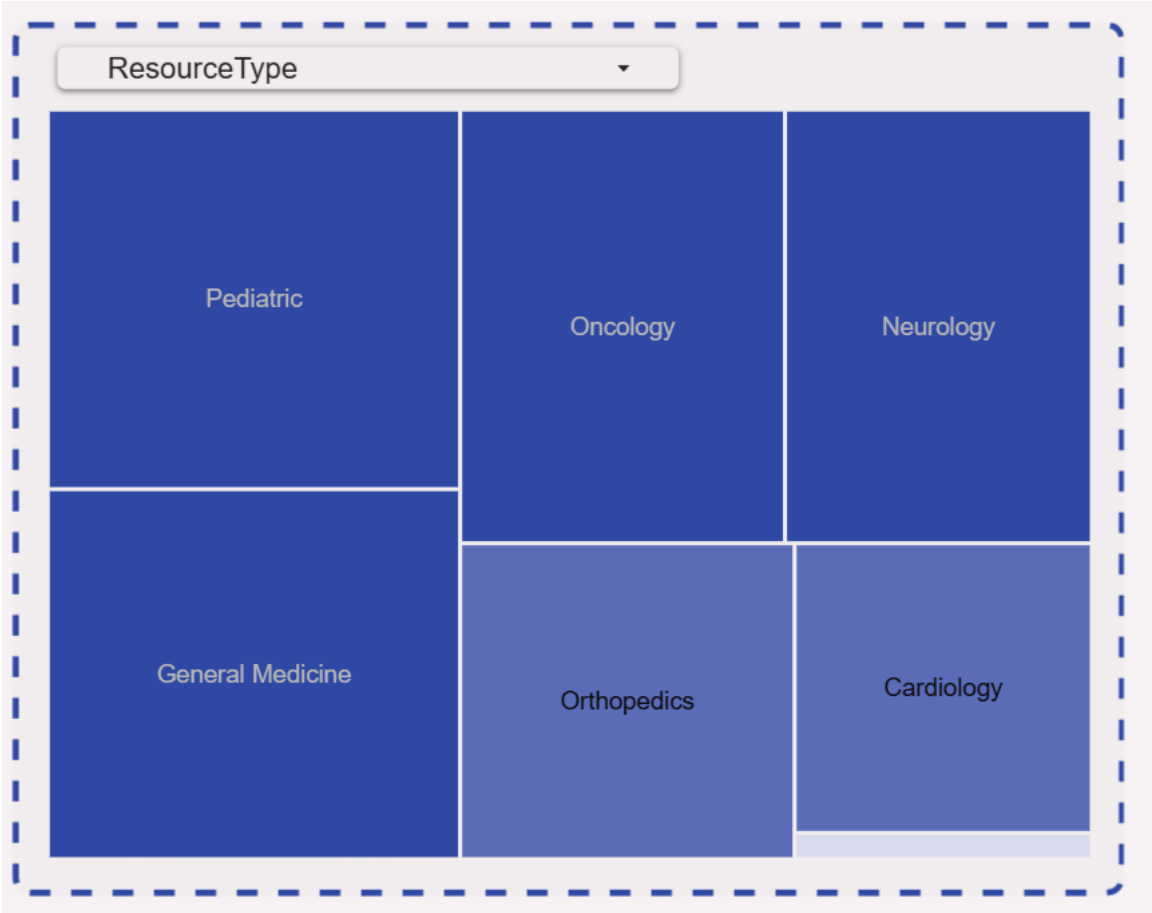
Data Visualization

Patient feedback visualized using a Pie Chart:



Data Visualization

Heat Map showing the efficiency of departments:



Risk Management Plan

Risks identified in the risk register:

Risk ID	Risk Description	Category	Likelihood	Impact	Severity	Mitigation Strategy
OR-1	Prolonged patient waiting time due to uneven resource allocation	Operational	High	High	High	Implement dynamic staffing models, redistribute resources from underutilized departments, optimize scheduling workflows
OR-2	Imbalance in resource utilization across departments	Operational	High	Medium	High	Periodic workload review, cross-department resource sharing, capacity planning
OR-3	Operational bottlenecks in Pediatric and General Medicine departments	Operational	Medium	High	High	Add peak-hour staffing, implement patient flow optimization, triage improvements
OR-4	Decline in patient satisfaction due to service delays	Operational	Medium	High	High	Continuous monitoring of wait times, service quality training, SLA enforcement
TR-1	Limited visualization capability (no native heatmap in Looker)	Technical	High	Low	Medium	Use treemap and conditional formatting as functional alternatives, document design decisions

Risk Management Plan

Risks identified in the risk register (2):

Risk ID	Risk Description	Category	Likelihood	Impact	Severity	Mitigation Strategy
TR-2	Data quality issues (missing values, inconsistent labels)	Technical	Medium	High	High	Implement automated data validation, standardized data cleaning pipelines, routine audits
TR-3	Dashboard performance degradation due to increasing data volume	Technical	Medium	Medium	Medium	Data aggregation, incremental refresh, query optimization, archival strategies
TR-4	Unauthorized access to sensitive healthcare data	Technical	Low	High	Medium	Role-based access control (RBAC), data anonymization, compliance audits
SR-1	Resistance to data-driven decision-making culture	Stakeholder	Medium	Medium	Medium	Conduct training sessions, leadership advocacy, demonstrate quick wins
SR-2	Misinterpretation of dashboard metrics by users	Stakeholder	Medium	Medium	Medium	User training, embedded dashboard guidance, KPI documentation

Risk Management Plan

Risks identified in the risk register (3):

Risk ID	Risk Description	Category	Likelihood	Impact	Severity	Mitigation Strategy
SR-3	Conflict between departments during resource redistribution	Stakeholder	Medium	High	High	Transparent communication, performance-aligned incentives, executive sponsorship
SR-4	Executive expectation gap regarding predictive capabilities	Stakeholder	Low	Medium	Low	Clear project scoping, roadmap presentation, phased BI development plan

Risk Management Plan

Risks categorized based on the Risk Assessment Matrix:

Likelihood/Impact	Low Impact	Medium Impact	High Impact
High Likelihood	Medium Risk	High Risk	High Risk
Medium Likelihood	Low Risk	Medium Risk	High Risk
Low Likelihood	Low Risk	Low Risk	Medium Risk

Risk Management Plan

Elements identified in the SWOT analysis:

Strengths

- Integrated multi-source operational analytics
- Clear KPI visibility for decision support
- Interactive, user-friendly dashboard design
- Visualizations enable fast issue detection

Weaknesses

- High average patient waiting time (~44.6 min)
- Imbalanced resource utilization across departments
- Limited predictive/advanced analytics capability
- Insights depend heavily on data quality

Opportunities

- Predictive demand & capacity planning
- Cost optimization through better resource use
- Improved patient experience & service flow
- Expansion toward digital healthcare ecosystem

Threats

- Fluctuating patient demand and surge risk
- Regulatory and data privacy compliance risks
- Staff workload concentration & burnout
- Dependency on technology and system reliability

Risk Management Plan

Key insights from the Risk Management Plan:

- Highest risks relate to **system integration, data quality, and user adoption**
- Resource and schedule constraints may impact delivery timeline and service continuity
- Regulatory and data privacy compliance are **critical risk areas**
- Change resistance and training gaps could slow system adoption
- Proactive monitoring and mitigation planning are essential to maintain project stability

Risk Mitigation Plan

Factors included in the Contingency Plan:

Risk ID	Contingency Plan
OR-1	Activate surge staffing, dynamic rescheduling, and rapid triage when waiting times exceed thresholds
OR-2	Reallocate staff and equipment across departments to rebalance resource utilization
OR-3	Deploy overflow units and fast-track workflows to relieve bottlenecks and stabilize patient flow
OR-4	Launch service recovery actions (communication, coaching, rapid complaint resolution) if satisfaction declines
TR-2	Trigger data audit and temporary manual reporting when data integrity issues occur
SR-3	Use executive mediation and pilot redistribution to resolve interdepartmental conflicts

Risk Mitigation Plan

Risks prioritized based on the Visual Risk Matrix:

Priority Level	Risk ID & Description	Rationale	Action Urgency
High	OR-1 – Prolonged patient waiting time	High likelihood and high impact directly affecting service efficiency and patient flow	Immediate mitigation required
High	OR-3 – Operational bottlenecks in key departments	Causes throughput delays and overload during peak demand	Immediate mitigation required
High	OR-4 – Decline in patient satisfaction	Directly impacts service quality and hospital reputation	Immediate mitigation required
High	TR-2 – Data quality and integrity issues	High impact on analytics accuracy and decision-making reliability	Immediate mitigation required
High	SR-3 – Interdepartmental conflict during resource redistribution	High impact on operational coordination and service continuity	Immediate mitigation required
Medium	OR-2 – Resource utilization imbalance	Moderate operational disruption but manageable through capacity planning	Scheduled mitigation

Risk Mitigation Plan

Risks prioritized based on the Visual Risk Matrix (2):

Priority Level	Risk ID & Description	Rationale	Action Urgency
Medium	TR-3 – Dashboard performance degradation	May affect system responsiveness but not core operations	Scheduled mitigation
Medium	TR-4 – Unauthorized data access risk	Security risk with serious impact but lower likelihood	Preventive controls required
Medium	SR-1 – Resistance to data-driven culture	May slow adoption and reduce system effectiveness	Continuous management
Medium	SR-2 – Misinterpretation of dashboard metrics	Can affect decision quality if unmanaged	Continuous training
Low	SR-4 – Executive expectation gap	Low likelihood and manageable through communication	Monitor only

Risk Mitigation Plan

Key insights from the Risk Mitigation Plan:

- Focus mitigation on **operational bottlenecks, waiting time, and data quality**, the highest-impact risks
- Use **dynamic staffing, resource reallocation, and flow optimization** to stabilize service performance
- Strengthen **data validation, security controls, and system monitoring** to ensure reliable decision-making
- Address **user adoption and cross-department coordination** through training, communication, and governance
- Apply **continuous risk monitoring** to detect early warning signals and trigger timely corrective actions

A photograph of a laptop on a wooden table in a cafe. The laptop screen displays a blue network diagram with nodes and connecting lines. A white mug is on the table to the right of the laptop. The background is a blurred cafe interior with warm, bokeh lights.

Findings and Recommendations

Key Findings

1. Appointment demand is uneven, with **peak overload in morning and evening** and underutilized midday capacity
2. Average patient waiting time remains **high (~44–45 min)**, indicating operational inefficiencies
3. Resource utilization is **imbalanced across departments**, creating bottlenecks in high-demand units
4. Data quality and system integration are **critical enablers** for reliable analytics and decision-making
5. Process fragmentation and manual coordination contribute to **delays and reduced service efficiency**
6. Targeted digital integration and operational optimization present **significant improvement potential**

Key Recommendations

1. Implement integrated digital appointment, check-in, and patient data systems
2. Optimize scheduling and staffing to balance peak and low-demand periods
3. Improve resource allocation across departments to reduce bottlenecks
4. Strengthen data quality, validation, and system reliability controls
5. Enhance interdepartmental coordination and workflow automation
6. Establish KPI-driven monitoring for continuous performance improvement

Conclusion

Provide a summary of observations in 3–5 bullet points.

1. Appointment demand is uneven, with peak overload in morning/evening and underutilized midday capacity
2. Average patient waiting time remains high, indicating process and capacity inefficiencies
3. Resource utilization is imbalanced across departments, creating localized bottlenecks
4. Data quality and system integration gaps limit decision accuracy and operational visibility
5. Manual coordination and fragmented workflows contribute to delays and reduced service efficiency